

# Data Flow Diagram

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## Level 0 — Context Diagram

User --> [Smart Lender System] --> Prediction Result --> User

## Level 1 — Detailed Data Flow

- 1. User enters applicant details into the HTML form (index.html).
- 2. Form data is sent via POST request to the Flask backend (/predict route).
- 3. Backend validates and transforms input into the feature format expected by the model.
- 4. The pre-trained ML model (model.pkl) receives the feature vector and returns a prediction.
- 5. Backend formats the prediction into a human-readable response.
- 6. Result is rendered on result.html and returned to the user's browser.

## Data Stores

- Training Dataset (CSV) — used only during the offline model training phase (Epic 1-4).
- Trained Model File (model.pkl) — used at runtime for predictions.

## External Entities

- Applicant / End User — provides input data, receives prediction.

## Processes

- P1: Collect User Input
- P2: Preprocess Input (encoding/scaling consistent with training)
- P3: Run Model Inference
- P4: Generate and Display Result